

DESHAN CHATHURANGA

Full Stack Software Engineer

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PROFILE

Data-focused engineer with nearly 4 years of experience building scalable backend systems and end-to-end data pipelines. Skilled in Python, SQL, and Java (Spring Boot), with hands-on experience across the data lifecycle from pipeline design to deployment. Built GridSight UK, a production-grade probabilistic forecasting platform for the National Grid using a medallion data pipeline architecture, the top-marked project in an MSc Data Science cohort (Distinction-level, University of Hertfordshire). Also bring strong full-stack expertise (Angular, React) and a track record of delivering complex API integrations across high-concurrency environments. Right to work in the UK via the Graduate Visa route; applying for PSW in December 2026.

EXPERIENCE & EDUCATION TIMELINE

Full-time Postgraduate Student - MSc Data Science, University of Hertfordshire, UK September 2025 - September 2026

- Maintained a Distinction-level standing while delivering the GridSight UK AI forecasting project as the primary research output, which achieved the highest mark in the cohort (see Projects section).
- **Modules:** Machine Learning, Neural Networks, Applied Data Science, and Data Mining, directly applied to production-grade ML pipelines.

Senior Software Engineer | Codegen International (Pvt) Ltd

May 2025 - September 2025

- Architected and led the end-to-end H2H (Host-to-Host) full-stack API integration with HotelBeds, automating high-volume data exchange and enhancing real-time booking synchronisation for 3 enterprise clients.
- Optimised backend data processing pipelines in Java and Spring Boot, achieving 100% successful production rollouts across large-scale client go-live phases and reducing average error rate by 25%.
- Engineered complex data models using Oracle DB to ensure referential integrity and efficient retrieval across distributed microservices handling 50k+ daily transactions.
- Managed containerised environments via Docker and Kubernetes, ensuring high availability (99.9% uptime SLO) and horizontal scalability in high-concurrency production environments.
- **Technologies:** Java, Spring Boot, Angular, Docker, Kubernetes, Oracle DB, CI/CD, ORM, REST APIs

Software Engineer | Codegen International (Pvt) Ltd

March 2023 - May 2025

- Modernised the Travelbooker SaaS platform serving 10+ enterprise travel operators by integrating Travelport GDS and payment gateways, improving booking throughput by 30%.
- Developed an auto-ticketing feature for Amadeus flight bookings, reducing manual processing time by 40% by delivering a complex end-to-end integration within a 10-day sprint.
- Implemented CI/CD pipelines (GitHub Actions) and ORM frameworks to standardise delivery workflows, cutting deployment lead times by 50% and eliminating environment drift.
- Collaborated in a 5-engineer Agile Scrum team, contributing to sprint planning, code reviews, and technical documentation across Spring Boot and Angular feature deliveries.
- **Technologies:** Java, Spring Boot, Angular, Docker, Kubernetes, Oracle DB, CI/CD, ORM, REST APIs

Trainee Software Engineer | Scienter Technologies (Pvt) Ltd

January 2022 - June 2022

- Accelerated mobile application delivery for Pizza Hut and Taco Bell (Sri Lanka) by contributing to full-cycle cross-platform feature development using React Native.
- Improved LuckeWallet app navigation reliability by integrating Google Maps API and Redux state management, reducing reported navigation errors by 30%.
- **Technologies:** React Native, Redux, Google Maps API

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript
- **Backend:** Spring Boot, Node.js, Django, REST & SOAP APIs
- **Frontend:** Angular, React.js, React Native, Redux, HTML/CSS
- **Databases:** Oracle DB, PostgreSQL, MySQL, MongoDB, Firebase
- **MLOps & Data:** MLflow, DVC, Optuna, DuckDB, Evidently AI, FastAPI
- **Cloud & DevOps:** AWS, Docker, Kubernetes, CI/CD, Git, Redis
- **ML & AI:** TensorFlow, PyTorch, Scikit-Learn, NLTK, LightGBM, PySpark
- **Methodologies:** Agile (Scrum), TDD, ORM, Microservices, Software Architecture

KEY PROJECTS

GridSight UK - AI-Based Solar Energy Forecasting | University of Hertfordshire, UK September 2025 - Present

- Designed and built an end-to-end probabilistic ML forecasting platform predicting UK solar generation across 6h, 12h, and 24h horizons, supporting National Grid renewable energy planning.
- Applied a Bronze→Silver→Gold medallion data pipeline (Parquet) with rigorous UTC alignment, DST exclusion, and 30-minute interval reindexing across PV_Live, NESO, and Open-Meteo sources.
- Developed three competing model families for probabilistic forecasting: a standalone LSTM-Q recurrent network (PyTorch), an ensemble Stacking Model combining tabular LightGBM-Q and sequential TCN-Q with a Linear-Q meta-learner (5-fold walk-forward CV), and an Amazon Chronos-2 time-series foundation model fine-tuned via LoRA (PEFT).
- Achieved KPI targets: nMAE ≤ 4.0%, Skill Score > 0.30 at 24h, and PICP within 0.78–0.82 evaluated using Diebold-Mariano significance tests.
- Deployed a production-grade forecasting stack containerized via multi-stage Docker Compose (FastAPI backend + React/Vite frontend) on Render; established GitHub Actions CI/CD with automated DVC artifact pulling, unit testing (pytest), and weekly Champion-vs-Challenger model retraining monitored by Evidently AI for data drift.
- Technologies:** Python, PyTorch, LightGBM, Hugging Face (PEFT/LoRA), FastAPI, React, Docker, MLOps, Time-Series Forecasting.

Automated RAG Pipeline & Policy Data Platform June 2026 - July 2026

- Engineered an ETL pipeline to ingest unstructured PDF reports, utilizing sliding-window chunking and batch embeddings (nomic-embed-text) to populate a persistent vector database.
- Implemented an automated data quality validation suite enforcing strict JSON schemas and leveraging cross-model checks to audit output faithfulness and consistency.
- Technologies:** Python, Streamlit, Ollama, Plotly, PyPDF, NumPy, JSON Schema

Intelligent Candidate Ranking System | University of Moratuwa, Sri Lanka June 2022 - June 2023

- Developed an end-to-end ML pipeline for automated candidate ranking, performing feature extraction from resumes and LinkedIn profiles using NLP techniques (Python, NLTK).
- Developed and tuned predictive models with Scikit-learn; architected a hybrid storage solution with MySQL (structured metadata) and MongoDB (document store).
- Technologies:** Python, NLTK, Django, Scikit-learn, MySQL, MongoDB

Leave Approval & Inventory Management System July 2020 - July 2021

- Planned an RBAC system managing hierarchical approval workflows ensuring data security and organizational compliance across departments.
- Built automated email and system notifications via Spring Boot to reduce manual processing delays.
- Technologies:** React.js, Spring Boot, MySQL

EDUCATION

MSc Data Science | University of Hertfordshire, UK September 2025 – July 2026 (Expected)
Current Grade: First Class

Major: Machine Learning, Neural Networks, Applied Data Science Minor: Data Mining

BSc (Hons) Information Technology | University of Moratuwa, Sri Lanka November 2018 - July 2023
Grade: Second Class Upper Division

Major: Software Systems & IT Minor: Machine Learning & Artificial Intelligence

CERTIFICATIONS & ACHIEVEMENTS

- AWS Fundamentals - Coursera
- Version Control with Git - Atlassian
- Scrum Foundation Professional Certificate - CertiProf
- Microsoft Cloud Skills Challenge Winner (2021) - IEEE Student Branch, Sri Lanka
- Merit Team - Home Alone Hackathon (2020)
- Spirit of Service Award - Rotaract Club, University of Moratuwa (2019–2020)

SOFT SKILLS

- Strong problem-solving and analytical skills with focus on performance and scalability
- Effective collaboration in Agile and cross-functional teams
- Quick learner with the ability to adopt and apply new technologies
- High attention to detail with strong ownership and accountability for deliverables
- Excellent communication and stakeholder collaboration skills